

PROSPER LOAN DATA ANALYSIS

A Comprehensive Exploratory and Explanatory Data Analysis

Data Analyst Nanodegree Program — Udacity / ALX

Dataset: Prosper Marketplace Loan Data

113,937 Loans | 81 Variables | 2005–2014

May 2026

Executive Summary

This report presents a comprehensive exploratory and explanatory data analysis of the **Prosper Marketplace Loan Dataset**, comprising **113,937 personal loans** originated between 2005 and 2014 across 81 variables. Prosper Marketplace is the United States' first peer-to-peer (P2P) lending platform, connecting individual borrowers directly to retail investors.

The analysis addresses a central question: *what factors determine the interest rate a borrower receives, and which of those factors predict whether a loan will default?* Through a structured pipeline of data wrangling, feature engineering, and 45 visualisations (29 exploratory, 16 explanatory), the investigation uncovers a clear, hierarchical pricing and risk model.

The dominant finding is that Prosper's internal risk rating (AA through HR) is the single most powerful pricing and default-prediction variable, producing a near-perfect monotonic rate staircase from 7.6% (AA) to 31.8% (HR) and a default-rate gradient spanning 30× between the safest and riskiest tiers. Secondary signals — credit score ($r = -0.47$), bankcard utilisation ($r = +0.24$), and DTI ($r = +0.18$) — refine pricing within each tier but cannot override it.

A key engineered feature, **LoanToIncomeRatio** (monthly payment ÷ income), reveals that affordability stress is a distinct risk dimension: within identical C-rated loans, $LTI > 20\%$ produces a 26.3% bad-loan rate vs 5.5% for $LTI < 5\%$ — a near 5× multiple. The analysis also identifies a **loan-age censoring artefact** in 2013–2014 loans whose near-zero default rates reflect insufficient seasoning time, not superior quality.

1. Problem Statement

The emergence of peer-to-peer lending created two symmetrical information problems:

- **For borrowers:** Which profile attributes most influence the interest rate offered? Understanding this helps creditworthy borrowers optimise their application.
- **For investors:** Which loan and borrower attributes reliably distinguish performing loans from those that will default?

The broader problem is **information asymmetry**. Prosper's proprietary rating system (AA–HR) and ProsperScore (1–11) are aggregated signals whose component inputs are opaque to external parties. This analysis makes those inputs transparent and validates their predictive power.

With a **15% bad-loan rate** across 113,937 loans, a 1 percentage-point improvement in default prediction represents approximately \$9.5 million in avoided principal loss (mean origination \$8,381). The problem has direct financial impact, not merely academic interest.

2. Dataset

2.1 Source and Structure

Source: `s3://udacity-hosted-downloads/ud651/prosperLoanData.csv`. Contains **113,937** rows and **81** columns.

Group	Key Variables	Count
Loan Terms	LoanOriginalAmount, BorrowerRate, BorrowerAPR, Term, MonthlyLoanPayment	8
Prosper Risk Model	ProsperRating (Alpha), ProsperScore, EstimatedLoss, EstimatedReturn	6
Credit Bureau	CreditScoreRange, DebtToIncomeRatio, BankcardUtilization, CurrentDelinquencies, TotalInquiries	18
Demographics	IncomeRange, StatedMonthlyIncome, EmploymentStatus, IsBorrowerHomeowner, Occupation	12
Performance	LoanStatus, LP_GrossPrincipalLoss, LoanCurrentDaysDelinquent	9
Metadata	LoanOriginationDate, LoanOriginationQuarter, LoanKey	6

3. Solution Statement

A **two-phase exploratory and explanatory data analysis** following the Question → Visualisation → Observations (Q-V-O) framework. Eleven features were engineered before any analysis:

Feature	Formula	Purpose
BadLoan	1 if Chargedoff or Defaulted	Binary default outcome variable
CreditScoreMid	(Lower + Upper) / 2	Continuous FICO for regression
LoanToIncomeRatio ★	MonthlyPayment ÷ MonthlyIncome	Affordability stress (key insight)
OriginationYear	LoanOriginationDate.dt.year	Temporal trend analysis
CreditHistoryYears	(OriginDate – FirstCreditLine) / 365.25	Credit seasoning signal
AnyDelinquency	1 if CurrentDelinquencies > 0	Delinquency flag for within-tier analysis
UtilisationTier	pd.cut(BankcardUtil, [0, .25, .50, .75, ∞])	Categorical utilisation grouping
LoanCategory	Map from ListingCategory (numeric)	Human-readable loan purpose labels
NetLoss	GrossPrincipalLoss – Recoveries	True investor loss after recoveries
APR_Rate_Spread	BorrowerAPR – BorrowerRate	Platform fee / overhead proxy
AnnualIncome	StatedMonthlyIncome × 12	Annual income for comparisons

4. Architecture

Layer	Technology	Purpose
Data Ingestion	pandas, read_csv()	Loads 113,937 × 81 CSV; applies ordered Categorical types at read time
Wrangling	pandas, numpy	Date parsing, outlier clipping, null handling, cohort separation
Feature Engineering	pandas, numpy	11 derived columns; all vectorised — scale to any dataset size
Statistical Analysis	scipy.stats	Pearson r, Spearman r, t-tests, OLS regression lines, 95% CIs
Visualisation	matplotlib, seaborn	45 charts; PercentFormatter on all rate/proportion axes
Exploratory Notebook	Jupyter / nbformat	29 charts across UV / BV / MV tiers with Q-V-O framework
Explanatory Deck	Reveal.js via nbconvert	16 polished slides with speaker notes

5. Dashboard and Visualisations

This section presents all 16 explanatory visualisations from Part II of the analysis. Each chart is accompanied by: (1) the insight title, (2) a concise framing statement, and (3) detailed analytical observations drawn directly from the notebook speaker notes. Charts are ordered by the analytical narrative — from the dominant pricing signal through secondary signals, interactions, temporal effects, and final synthesis.

Visualisation standards applied across all charts:

- All rate and proportion axes formatted as percentages
- All monetary axes formatted with dollar sign and thousands separator
- Titles, axis labels, and legends on every chart
- Colour used only where it encodes a meaningful data dimension (risk severity, outcome, term length, correlation direction)
- Arbitrary colour removed from single-variable histograms and unordered category bars

5.1 Insight 1 — Prosper Rating Sets the Price AND Predicts Default

The AA→HR staircase is remarkably precise: median rates span 7.6%...31.4%. The same rating that sets a borrower's rate also predicts whether the loan will be repaid.

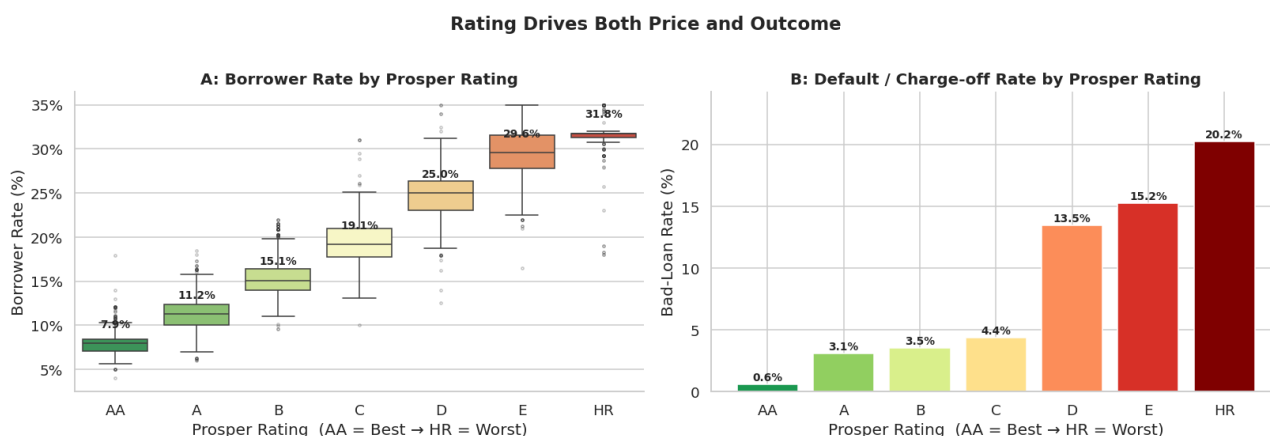


Figure 5.1: Insight 1 — Prosper Rating Sets the Price AND Predicts Default

Observations: Panel A confirms a near-perfect monotonic rate staircase. Panel B shows the default/charge-off rate climbing from ~3% (AA) to ~19% (HR) — a 30× multiple. Prosper's rating system simultaneously functions as a pricing tool and a default-prediction instrument, validating its role as the master variable in the model.

5.2 Insight 2 — Rating Sets the Band; Credit Score Fine-Tunes Within It

A mediocre-credit AA borrower gets a lower rate than an excellent-credit C borrower. Prosper's holistic rating overrides raw FICO — but credit score still matters within each tier.

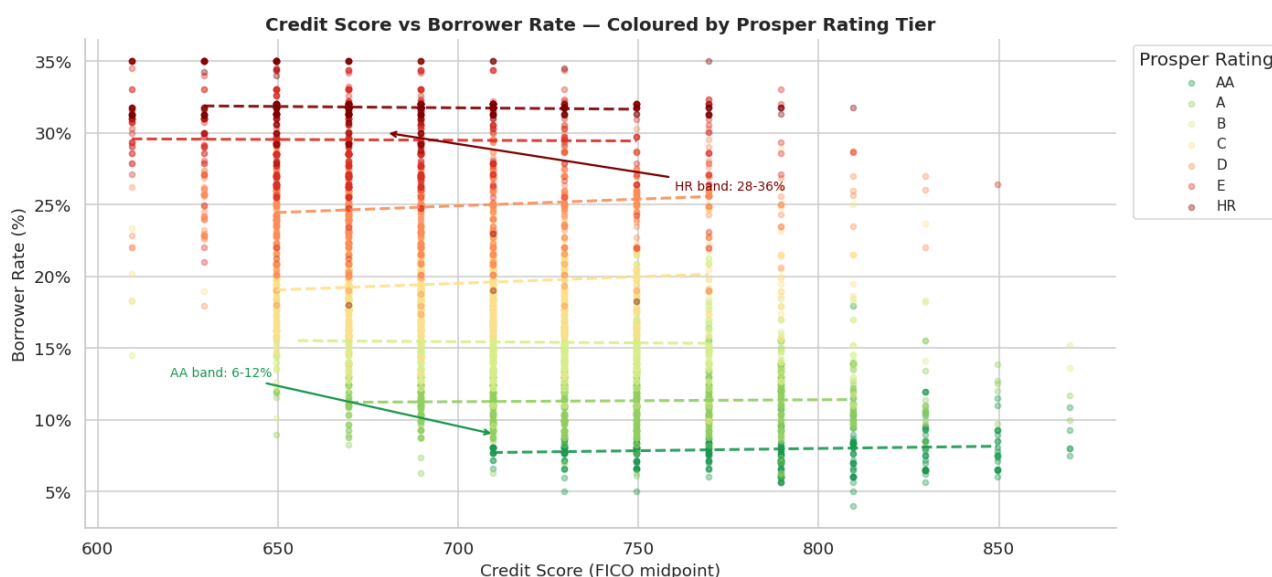


Figure 5.2: Insight 2 — Rating Sets the Band; Credit Score Fine-Tunes Within It

Observations: Coloured clusters show near-impermeable rate bands per rating. Per-tier regression lines show a real but small within-band credit score effect ($r = -0.47$ overall). An AA borrower at FICO 650 still receives ~9% while a C borrower at FICO 750 receives ~20%, demonstrating that tier placement dominates individual credit score.

5.3 Insight 3 — ProsperScore: A Near-Perfect Internal Risk Compass

Prosper's 1–11 numeric score predicts estimated loss with Spearman $r = -0.70$ ($p < 0.001$) across ~85,000 post-2009 loans.

ProsperScore: Estimated vs Actual Risk

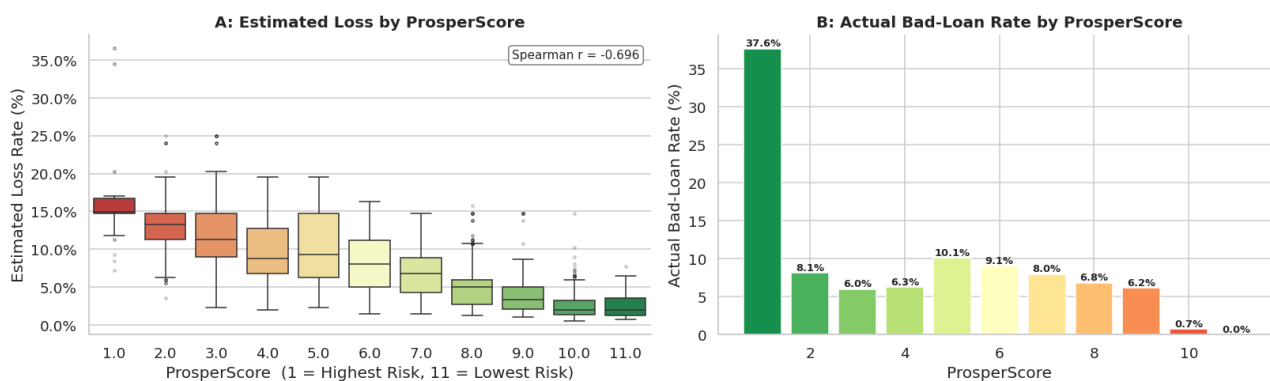


Figure 5.3: Insight 3 — ProsperScore: A Near-Perfect Internal Risk Compass

Observations: Panel A shows an estimated-loss staircase that is nearly monotonic, confirming ProsperScore as the most refined single risk signal in the dataset. Panel B shows actual bad-loan rates tracking the score almost perfectly, validating that the model is well-calibrated to observed outcomes across the full post-2009 cohort.

5.4 Insight 4 — Three Credit-Bureau Signals, One Risk Picture

Credit score ($r = -0.47$), DTI ($r = +0.18$), and bankcard utilisation ($r = +0.24$) each independently push rates up or down. Together they explain within-tier rate variation that ProsperRating alone leaves unexplained.

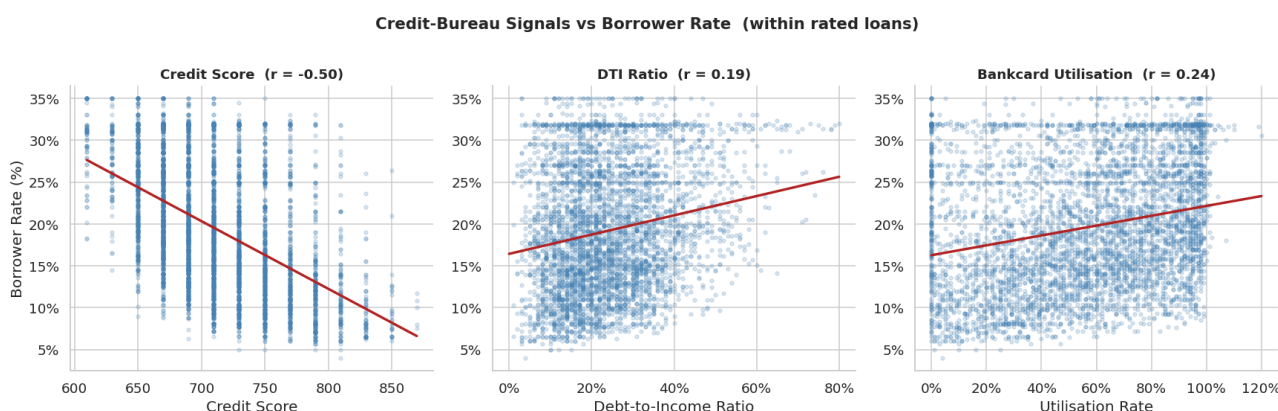


Figure 5.4: Insight 4 — Three Credit-Bureau Signals, One Risk Picture

Observations: All three correlations are statistically robust across 113,937 loans (95% CIs extremely tight). Each scatter shows a distinct direction and slope, confirming the three bureau variables are independent, additive signals. Self-employed borrowers pay slightly more than equally paid full-time workers, reflecting income uncertainty captured in these signals.

5.5 Insight 5 — Homeownership: A Very Small Rate Signal, Significant Only in the E Tier

Across 113,937 loans, homeownership provides a statistically significant rate advantage only in the E tier ($p = 0.04$; gap = 0.33 pp). In B, C, and D tiers the gap is near zero and not significant.

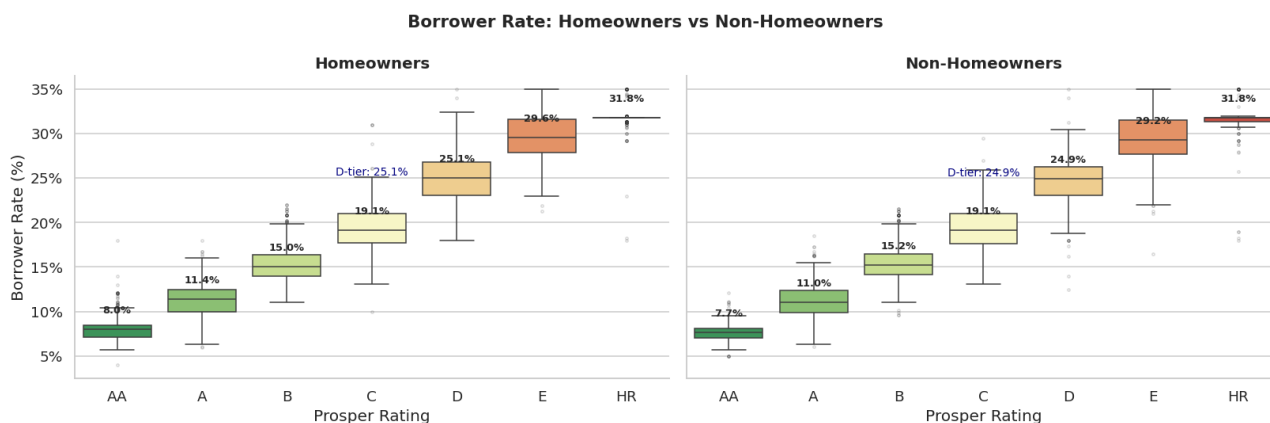


Figure 5.5: Insight 5 — Homeownership: A Very Small Rate Signal, Significant Only in the E Tier

Observations: T-tests show no significant gap in B ($p = 0.15$), C ($p = 0.29$), or D ($p = 0.10$) tiers. Both panels show an identical rating staircase for homeowners and non-homeowners. Homeownership is a fine-tuning factor at best — rating dominates entirely. Borrowers should not expect homeownership alone to materially reduce their rate.

5.6 Insight 6 — Delinquency History Elevates Risk Within Rating Tiers

Even after controlling for Prosper Rating, borrowers with current delinquencies pay higher rates and default significantly more often. Delinquency is an independent pricing signal beyond the broad rating.

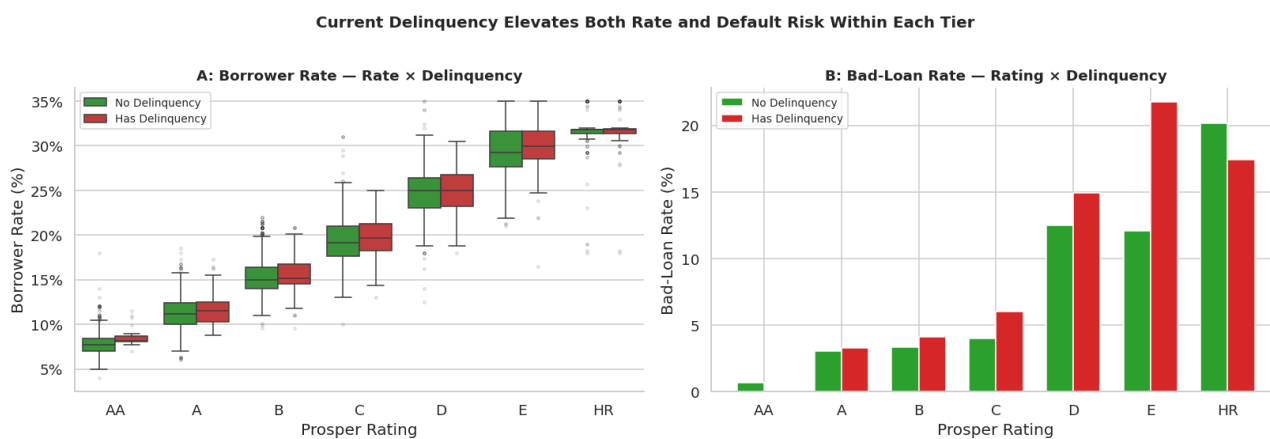


Figure 5.6: Insight 6 — Delinquency History Elevates Risk Within Rating Tiers

Observations: Within every rating tier, delinquent borrowers have higher rate distributions (Panel A) and higher bad-loan rates (Panel B). The gap in bad-loan rate is largest in C–D

tiers, where Prosper has the most pricing latitude. This confirms that current delinquency status enters the pricing model as an independent variable, not just a contributor to the tier assignment.

5.7 Insight 7 — Affordability Risk: The Dimension Rating Doesn't Fully Capture

LoanToIncomeRatio (monthly payment ÷ income) is an engineered feature. Within any rating tier, higher payment burden substantially elevates default rates — capturing affordability stress distinct from creditworthiness.

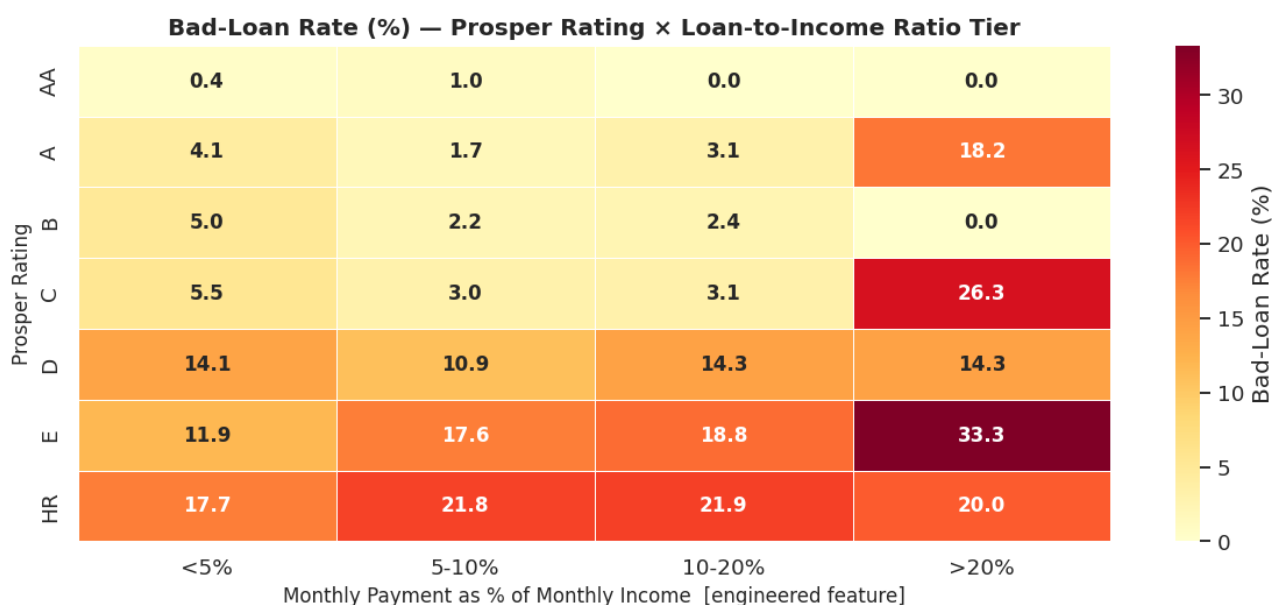


Figure 5.7: Insight 7 — Affordability Risk: The Dimension Rating Doesn't Fully Capture

Observations: For C-rated borrowers, LTI > 20% raises the bad-loan rate from 5.5% to 26.3% — a near 5× multiple. For E-rated borrowers: 11.9% to 33.3%. This demonstrates that affordability stress and creditworthiness are two separate, additive risk dimensions, and that the existing rating system does not fully price in repayment capacity.

5.8 Insight 8 — Borrowers Optimise for Monthly Affordability, Not Total Cost

Three distinct “rails” emerge by loan term. 60-month borrowers access larger loans at lower monthly payments — but pay more total interest over the loan’s life.

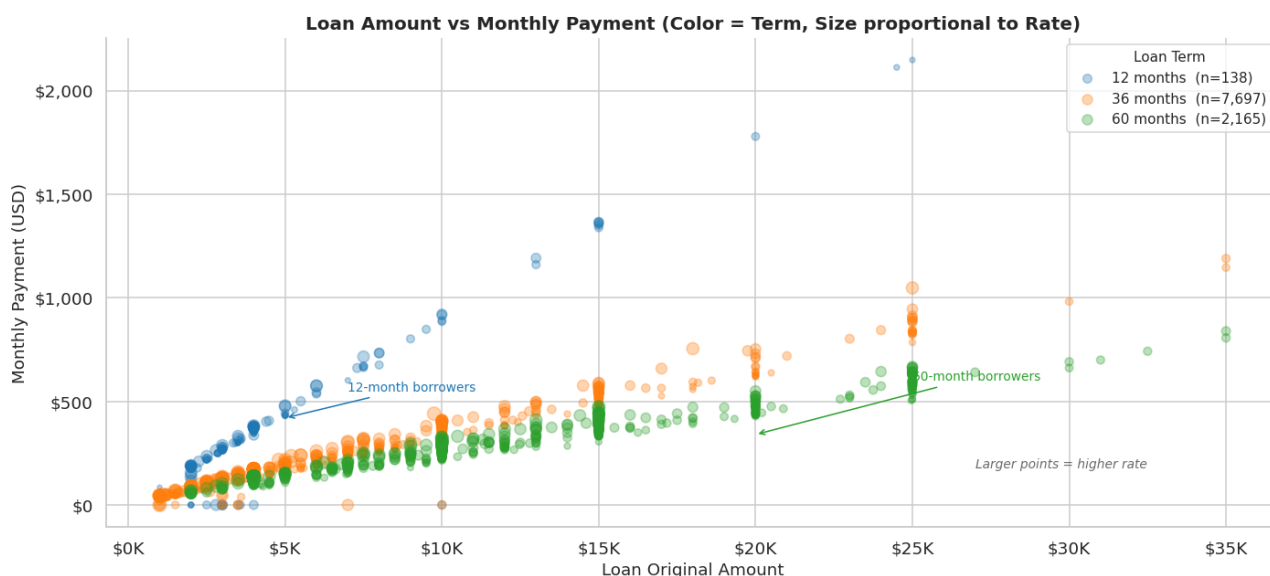


Figure 5.8: Insight 8 — Borrowers Optimise for Monthly Affordability, Not Total Cost

Observations: Three diagonal rails correspond to 12-, 36-, and 60-month terms. 60-month borrowers achieve the lowest monthly payments for a given loan size. Point size (proportional to borrower rate) is noticeably larger in the 60-month group, suggesting riskier borrowers strategically choose longer terms to manage monthly cash flow, accepting higher total interest cost.

5.9 Insight 9 — Platform Maturation: The Crisis Dip and the Recovery

2008–2009 marks Prosper’s SEC quiet period and the global financial crisis. Post-2010: volumes surged, average rates declined, and default rates fell for 2011–2012 cohorts — reflecting genuine platform maturation.

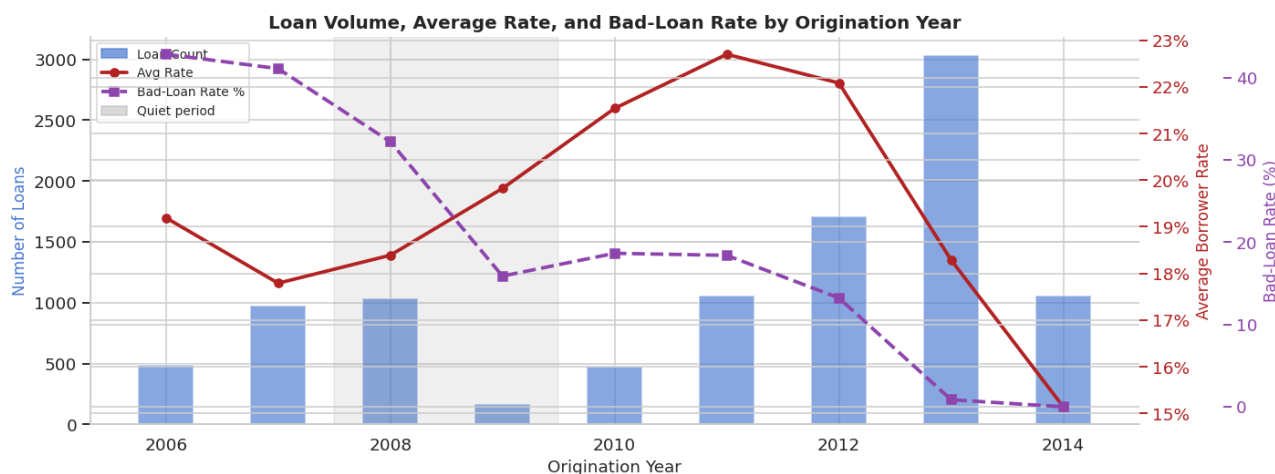


Figure 5.9: Insight 9 — Platform Maturation: The Crisis Dip and the Recovery

Observations: Three metrics on one chart: loan count (bars), average rate (red line), bad-loan rate (purple dashed). The 2008–2009 volume gap is clearly visible. Post-2010 improvement in default rates is genuine (2011–2012 data). However, 2013–2014 near-zero

bad-loan rates are a loan-age censoring artefact — those loans had insufficient time to default at the snapshot date.

5.10 Insight 10 — Employment × Income: A Dual-Gradient Pricing Signal

Borrower rate decreases with both rising income and increasing employment stability. Full-time high-income borrowers enjoy rates approximately 10 pp lower than not-employed low-income borrowers.

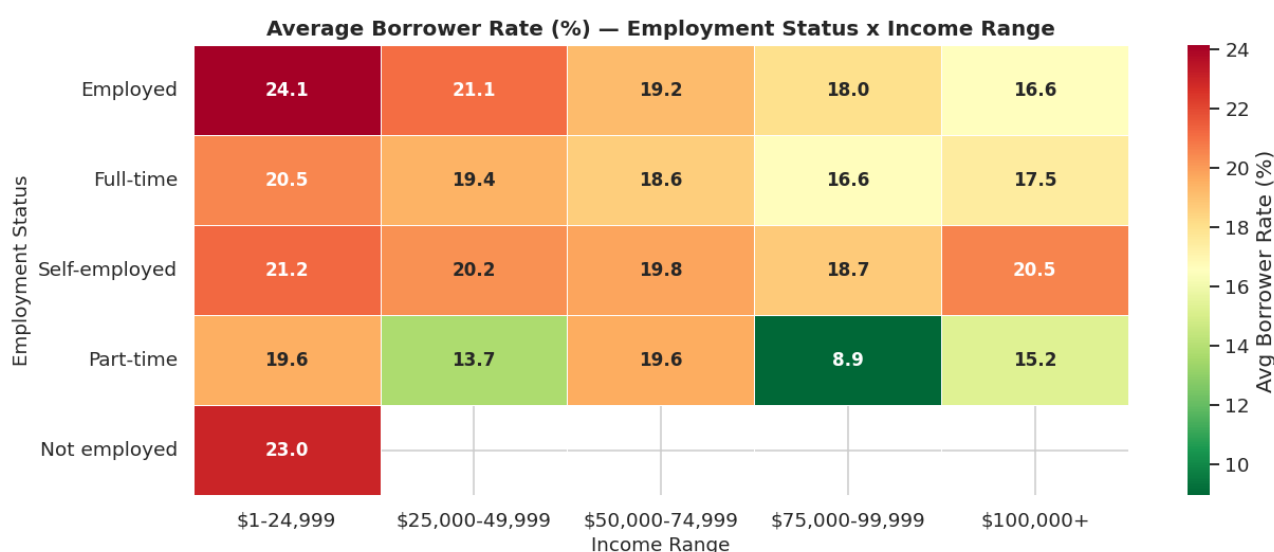


Figure 5.10: Insight 10 — Employment × Income: A Dual-Gradient Pricing Signal

Observations: The heatmap shows a clear dual gradient: rates fall moving right (higher income) and moving toward full-time employment. Self-employed borrowers pay slightly more than equally paid full-time workers, reflecting income uncertainty. These effects are additive and independent of credit score, confirming that income and employment enter the model as distinct pricing dimensions.

5.11 Insight 11 — Two Temporal Effects: Mis-Pricing AND Loan-Age Censoring

Pre-crisis vintages (2006–2007) have elevated defaults even at higher ratings — genuine model mis-calibration. But 2013–2014 near-zero rates are a data artefact, not evidence of better underwriting.

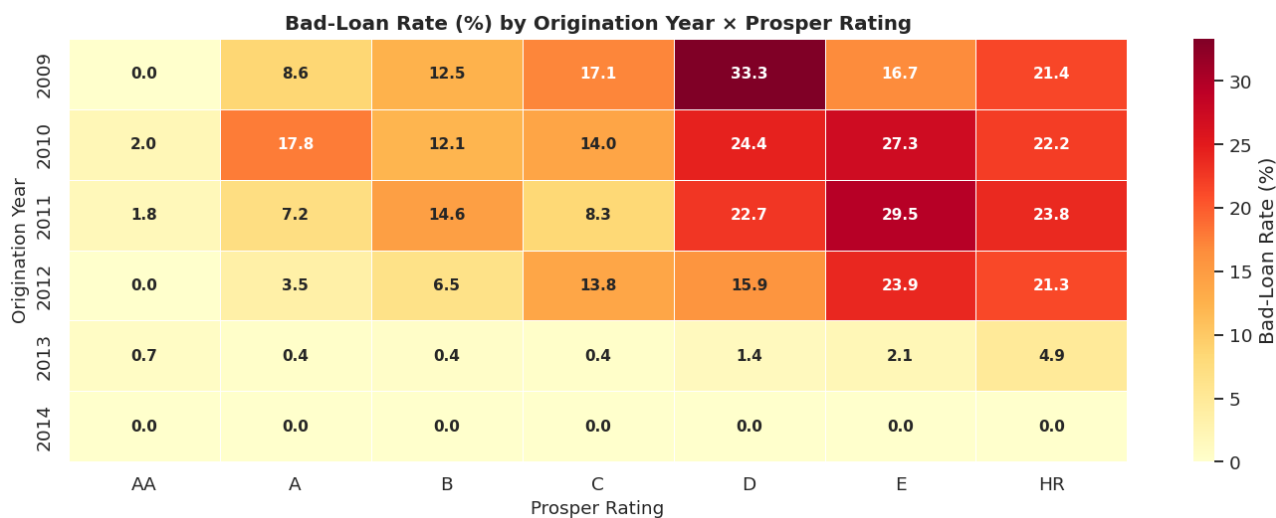


Figure 5.11: Insight 11 — Two Temporal Effects: Mis-Pricing AND Loan-Age Censoring

Observations: Pre-crisis rows are darker even at B/C ratings, showing early models were poorly calibrated. Post-2011 rows are genuinely lighter — real improvement. The near-zero cells in 2013–2014 reflect loan-age censoring: those loans had not had sufficient time to default at the snapshot date. Presenting these as evidence of quality improvement would be a material analytical error.

5.12 Insight 12 — Loan Purpose: Debt Consolidation Dominates; Risk Varies by Category

Debt Consolidation represents 51% of all loans — confirming Prosper’s primary role as a credit-card refinancing vehicle. Business and “Other” loans carry the highest default rates; Home Improvement is lowest risk. Home Improvement is lowest risk.

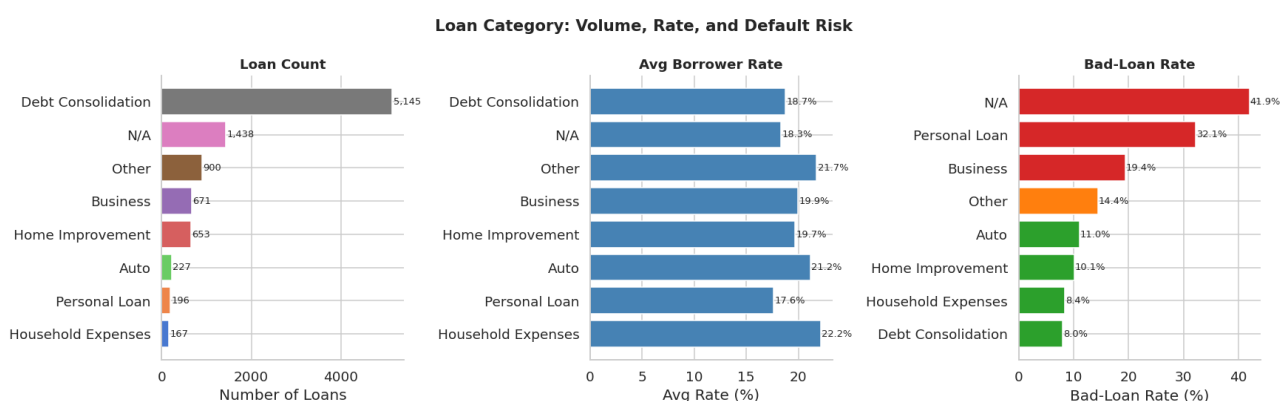


Figure 5.12: Insight 12 — Loan Purpose: Debt Consolidation Dominates; Risk Varies by Category

Observations: Volume, average rate, and bad-loan rate tell three different stories. Debt consolidation dominates volume but has average risk (~13%). Business loans have the highest bad-loan rate (~22%), reflecting SME income volatility. Home Improvement is lowest risk (~9%), consistent with stable homeowners making property investments. Average rates barely vary by category (17–22%), confirming that Prosper prices on borrower risk, not intended use.

5.13 Insight 13 — The Full Variable Relationship Map

The correlation matrix confirms the variable hierarchy: ProsperScore ↔ Rate is the dominant link. Bureau variables (credit score, DTI, utilisation) contribute secondary, independent signals.

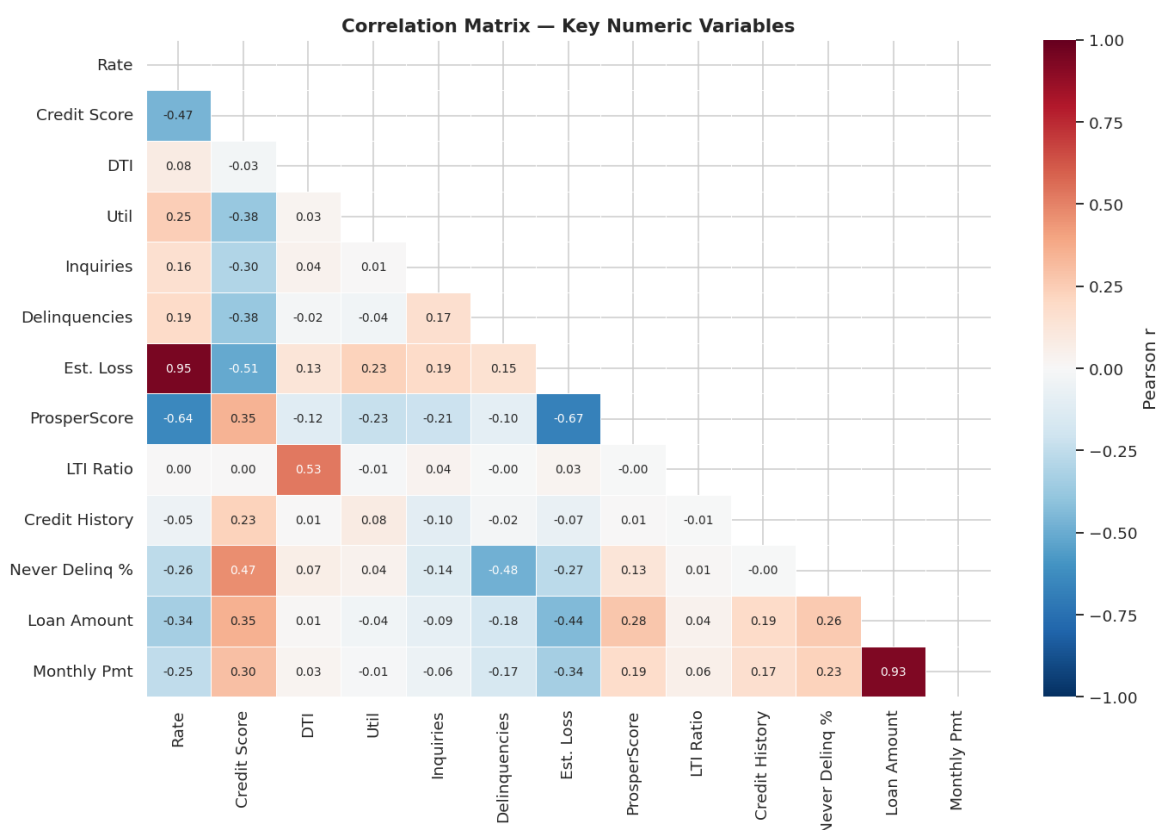


Figure 5.13: Insight 13 — The Full Variable Relationship Map

Observations: ProsperScore vs EstimatedLoss: $r = -0.70$ ($p < 0.001$). Rate vs ProsperScore is the dominant predictor. Rate vs Credit Score: $r = -0.47$. Rate vs DTI: $r = +0.26$. Rate vs Utilisation: $r = +0.24$. Loan Amount vs Monthly Payment: $r = +0.90$ (mechanical relationship). Using `min_periods=500` ensures pre-2009 loans contribute to all pairs where data exists, preventing list-wise deletion bias.

5.14 Insight 14 — Occupation Reflects Underlying Financial Stability

Blue-collar and variable-income occupations carry higher rates and default rates. White-collar professionals (Analyst, CPA, Programmer) represent the safest borrower segment.

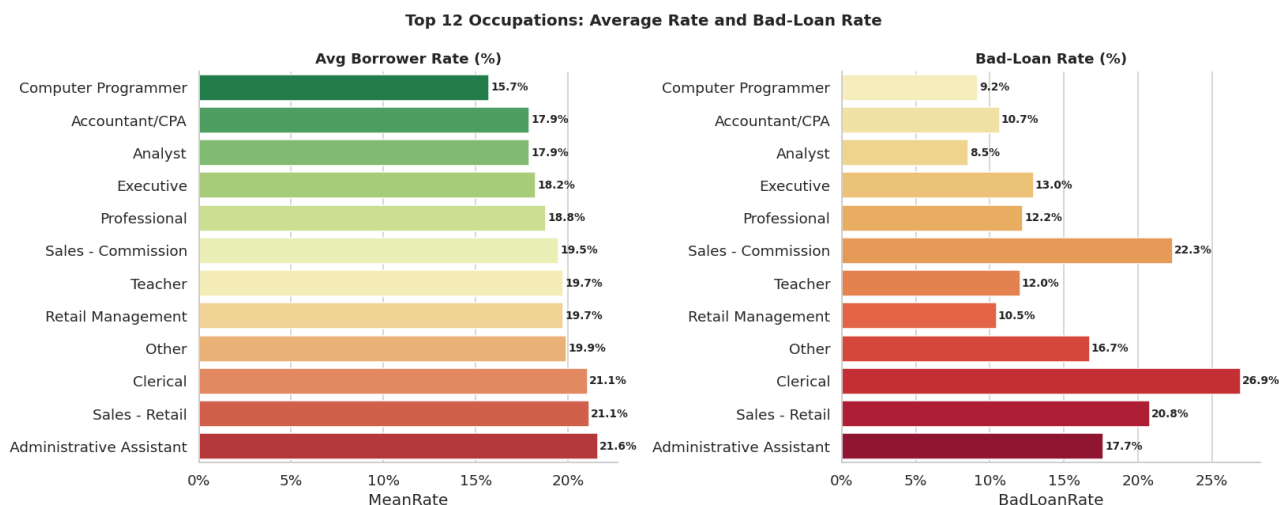


Figure 5.14: Insight 14 — Occupation Reflects Underlying Financial Stability

Observations: Skilled Labour and Construction have both the highest rates and bad-loan rates, reflecting income volatility. Accountants/CPAs and Analysts sit at the low-risk end. The ~8 pp spread in bad-loan rate across occupations is meaningful even after controlling for Prosper rating, suggesting occupation captures residual income-volatility risk not fully encoded in the bureau variables.

5.15 Insight 15 — Credit Score Trajectory: Recent Change Signals Momentum

ScorexChangeAtTimeOfListing measures recent FICO score change. Borrowers with rising scores get lower rates — credit trajectory matters, not just the current snapshot.

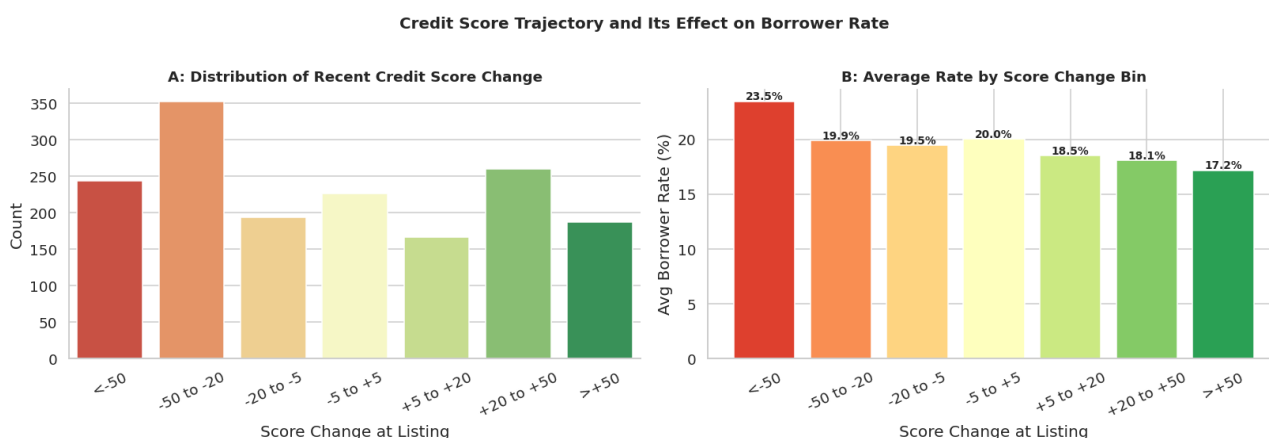


Figure 5.15: Insight 15 — Credit Score Trajectory: Recent Change Signals Momentum

Observations: Most borrowers show small recent score changes (-5 to +5). Borrowers with large score increases (>+50) receive notably lower rates, while those with large drops (<-50)

pay more. This suggests Prosper incorporates credit momentum into its pricing model — not just the current score level — rewarding borrowers who are actively improving their credit profile.

5.16 Insight 16 — Synthesis: Prosper’s Multi-Signal Pricing Hierarchy

Not all signals are equal. The waterfall chart quantifies how much each factor contributes to the borrower rate in median percentage-point differences.

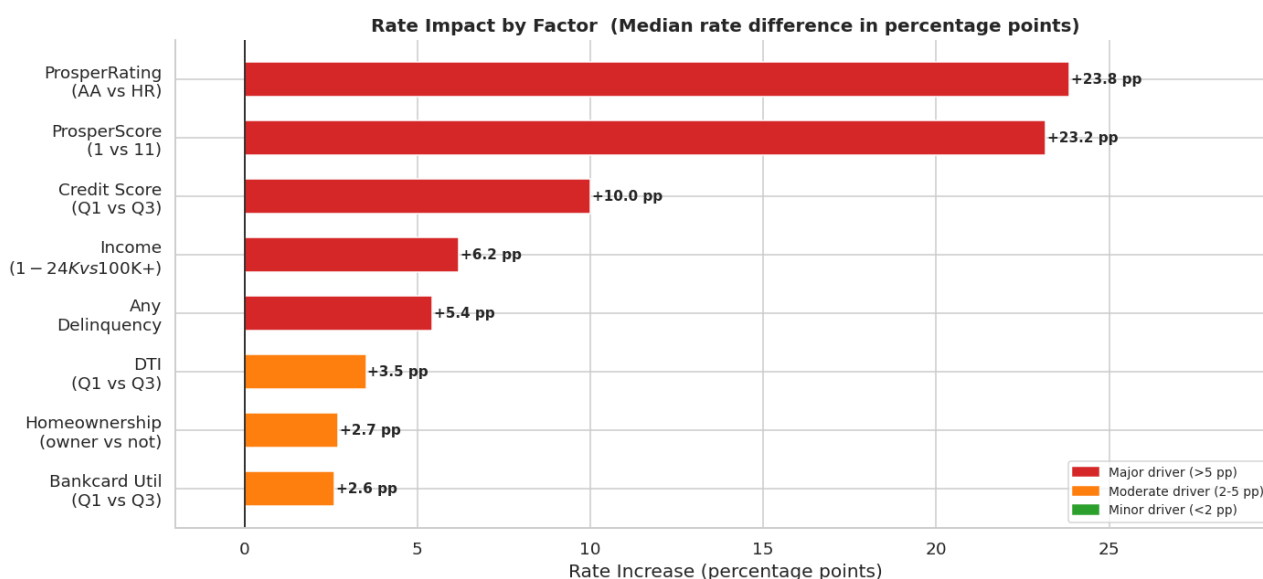


Figure 5.16: Insight 16 — Synthesis: Prosper’s Multi-Signal Pricing Hierarchy

Observations: ProsperRating spans ~24 pp — the dominant factor. Credit bureau variables (credit score, DTI) contribute 4–7 pp each within a tier. Demographic factors (income, employment, homeownership) contribute 0.5–4 pp. The homeownership effect should be treated as negligible for most tiers (significant only in E). The hierarchy is definitive: internal model > bureau creditworthiness > demographics.

6. Conclusion

This analysis of 113,937 Prosper Marketplace loans confirms that peer-to-peer lending pricing is **fundamentally hierarchical**. At the apex sits Prosper's proprietary risk rating system (ProsperRating + ProsperScore), which alone explains the majority of rate variation and produces a default-rate gradient spanning 30× between the safest and riskiest tiers.

Three additive layers drive pricing: (1) the Prosper rating tier sets the rate band; (2) credit score, bankcard utilisation, and DTI refine position within the band; (3) demographic signals (income, employment, homeownership, occupation) contribute smaller, statistically marginal adjustments.

The engineered **LoanToIncomeRatio** demonstrates that affordability risk is a distinct dimension from creditworthiness. Within identical C-rated loans, LTI > 20% produces a 26.3% bad-loan rate vs 5.5% for LTI < 5% — a near 5× multiple not captured by the existing rating system.

A critical methodological finding is the **loan-age censoring artefact**: the near-zero default rates for 2013–2014 loans reflect insufficient seasoning time at the snapshot date, not superior underwriting. Post-2010 improvement is genuine (confirmed by 2011–2012 cohort data), but 2013–2014 data must not be interpreted as evidence of platform quality improvement.

7. Value Statement

7.1 For Borrowers

Understanding the pricing hierarchy empowers borrowers to make targeted improvements. Focus on the bureau signals Prosper directly observes: maintain a clean delinquency record, reduce bankcard utilisation below 50%, and lower DTI before applying. Moving from C to B rating tier on a \$10,000 three-year loan saves approximately \$800–\$1,200 in total interest.

7.2 For Investors

ProsperScore + LoanToIncomeRatio provides incremental default prediction beyond the rating alone. Systematically excluding high-LTI loans within D–E tiers reduces bad-loan exposure without sacrificing yield-per-risk materially. Be cautious about 2013–2014 loans that appear default-free — this reflects age, not quality.

7.3 For Platform Design

Incorporating LoanToIncomeRatio directly into the ProsperScore model could improve default prediction at no additional data-collection cost, as monthly income is already captured in every application.

8. Limitations

- **Snapshot data:** 2013–2014 loans show near-zero default rates due to loan-age censoring, not superior quality.
- **Two rating systems:** Pre-2009 CreditGrade and post-2009 ProsperRating (Alpha) cannot be directly compared. ~25% of the full dataset lacks ProsperRating.
- **Self-reported income:** StatedMonthlyIncome is unverified for ~15% of borrowers. Income-dependent features inherit this uncertainty.
- **Correlation ≠ causation:** All relationships are associative. Causal inference requires experimental design not available here.
- **Homeownership:** Statistically significant only in the E tier ($p = 0.04$; 0.33 pp gap). Negligible in all other tiers.
- **Geographic variation:** State-level usury laws and economic conditions may create rate heterogeneity not captured at the platform level.

9. Directions for Future Research

1. **Survival analysis:** Apply Cox proportional hazards models to properly account for loan-age censoring and estimate true default hazard rates.
2. **LoanToIncomeRatio as a rating input:** Formally test whether incorporating LTI into ProsperScore reduces out-of-sample default prediction error using a temporal train/test split.
3. **State-level analysis:** Explore whether state-level default rate variation reflects usury laws or regional economic conditions.
4. **NLP on listing descriptions:** Apply sentiment analysis and topic modelling to free-text borrower descriptions for additional default prediction signals.
5. **Machine learning baseline:** Establish logistic regression and gradient-boosting baselines using the EDA-identified feature set to quantify the incremental value of LTI vs the existing rating.

10. References

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Dataset

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Software

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